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ORIGINAL RESEARCH

Real-time insect tracking and monitoring with computer vision and deep learning

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Abstract

Insects are declining in abundance and diversity, but their population trends remain uncertain as insects are difficult to monitor. Manual methods require substantial time investment in trapping and subsequent species identification. Camera trapping can alleviate some of the manual fieldwork, but the large quantities of image data are challenging to analyse. By embedding the image analyses into the recording process using computer vision techniques, it is possible to focus efforts on the most ecologically relevant image data. Here, we present an intelligent camera system, capable of detecting, tracking, and identifying individual insects *in situ*. We constructed the system from commercial off-the-shelf components and used deep learning open source software to perform species detection and classification. We present the Insect Classification and Tracking algorithm (ICT) that performs real-time classification and tracking at 0.33 frames per second. The system can upload summary data on the identity and movement track of insects to a server via the internet on a daily basis. We tested our system during the summer 2020 and detected 2994 insect tracks across 98 days. We achieved an average precision of 89% for correctly classified insect tracks of eight different species. This result was based on 504 manually verified tracks observed in videos during 10 days with varying insect activities. Using the track data, we could estimate the mean residence time for individual flower visiting insects within the field of view of the camera, and we were able to show a substantial variation in residence time among insect taxa. For honeybees, which were most abundant, residence time also varied through the season in relation to the plant species in bloom. Our proposed automated system showed promising results in non-destructive and real-time monitoring of insects and provides novel information about phenology, abundance, foraging behaviour, and movement ecology of flower visiting insects.

Introduction

Reports of declining abundance, diversity, and biomass of insects (Wagner et al., 2021) have sparked renewed interest in long-term monitoring (Didham et al., 2020). Traditionally, such monitoring has been done using active (e.g. sweep netting) or passive (e.g. pitfall or malaise trap) sampling techniques, where specimens are subsequently identified based on morphology (Montgomery et al., 2021) or molecular information (Amjad Bashir et al., 2014; Montgomery et al., 2021; O'Hara et al., 2008). For many taxa, long-term population census data are currently non-existent and more extensive monitoring is needed to decipher their population dynamics. Automated insect monitoring systems would therefore greatly advance our understanding of the spatial and taxonomical extent of insect declines, an important goal for pollination ecology (Høye et al., 2021; MacLeod et al., 2010).

So far, only few studies have tested image-based monitoring of insects with time-lapse cameras. Some studies have involved trapping insects in front of a camera and running detection and classification models on time-lapse images (Collett & Fisher, 2017; Eliopoulos et al., 2018; Xia et al., 2018). One study has described an adapted

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commercial camera for recording visits of bumblebees to a single flower, but did not actually demonstrate its use for long-term monitoring (Steen, 2017). The Rana system uses active motion vision to detect moving objects that meet criteria on target object size and shape (blob detection) and movement (automated tracking). Barlow et al. (2017) deployed several Rana units in the Rio Mesa reserve (Utah, USA) to monitor populations of native wildflower species to assess each species´ ability to support communities of pollinators. The units used a Raspberry Pi minicomputer, equipped with a low-resolution web-cam. A total of 1613 visits to the ten target plant species were recorded. The videos were manually evaluated for visitor identity, behaviour, and frequency requiring 96.5 person-hours for this study. An edge computing solution performing deep learning to automate the recognition and species classification into the recording process would greatly advance this technology.

Monitoring with cameras running high frame-rates facilitates the study of individual insect behavior and can improve the accuracy of automatic species classification as several images of the same individual are available for the identification, provided that the individual insects can be correctly tracked between frames. Automated video monitoring of insect pollinators in the field using direct manual observations is a time-consuming process and logistically limited by labour force and environmental conditions (Pegoraro et al., 2020). Continuous video monitoring systems that use low-cost commercial off-the-shelf recording devices are available. Many of these monitoring systems use handheld Digital Video Recorders (DVR) or weatherproof action cams (GoPro Hero 3) (Gilpin et al., 2017). Video monitoring has a high demand for power and storage space and is typically only a viable solution when grid power is available. Furthermore, the footage can be very time consuming to analyse subsequently without automated procedures.

Architectures of convolutional neural networks (CNNs) have provided particularly positive results in many areas of computer vision (Liu et al., 2017). CNNs use both pixel intensity values and spatial information about objects in the image. The technology is rapidly being adapted to ecological research (Christin et al., 2019), albeit mostly carried by the agricultural domain. Some studies have demonstrated the relevance and potential for applying CNNs in insect identification and monitoring both *in situ* and in lab settings (Høye et al., 2021). In image-based identification of insect specimens, CNNs have also proven very powerful (Ärje et al., 2020; Hansen et al., 2020). For detecting and tracking objects in real-time, computational speed is critical, and the open source framework You Only Look Once (YOLO), which is known to be particularly fast has gained strong support (Redmon et al., 2016). YOLO (Redmon et al., 2016) is a unified object detection framework for deep neural networks written in C and CUDA capable of running at 20 fps on a fast GPU.

In this paper, we present a novel method for automated real-time monitoring and classification of live individually tracked insects of great relevance to pollination ecology in particular and insect monitoring in general It is composed of off-the-shelf hardware and supports sufficient storage space and power supply to monitor insects over the entire growing season. We present a fully functional image-processing pipeline that utilizes deep learning for insect species classification and tracking. A small, power efficient computer, the Jetson Nano from NVIDIA (NVIDIA, 2021), running YOLO is used to detect and classify insect species based on images collected in real-time. The system is a low-cost solution integrated with open source code. We demonstrate its application by showing results from a full growing season and illustrate how tracking individuals increase classification accuracy and remove bias in abundance estimates caused by seasonal and taxonomical variation in residence time of individual insects on flowers.

Materials and Methods

The construction and selection of hardware components for the system and the software solution to identify and track insects in real-time are described in the following sections. Finally, an experiment and test of the insect monitoring system are presented.

Hardware solution

The primary components of the automated computer vision system were the computer from NVIDIA Jetson Nano Developer Kit (NVIDIA, 2021) that has a central processing unit (CPU) and a graphical processing unit (GPU) and a web camera, the Brio Ultra HD Pro Webcam (Logitech, 2021). An active cooling system was mounted on the computer to avoid overheating during warm summer days with continuous GPU processing. Images were stored on a 500 GB solid-state disk (SSD) drive. The computer was connected to the internet via a wireless network provided by a 4G mobile WiFi router. Data on the location and identity of the insects were transferred over the internet on a daily basis. A backup of all data was stored on the SSD and the system would work without internet access.

Figure 1 shows a view of the internal layout of the components of the system. The Jetson Nano was powered by 5 V and set to 5 W operation mode. With a camera and an SSD hard drive, the system consumes only 5–8 W

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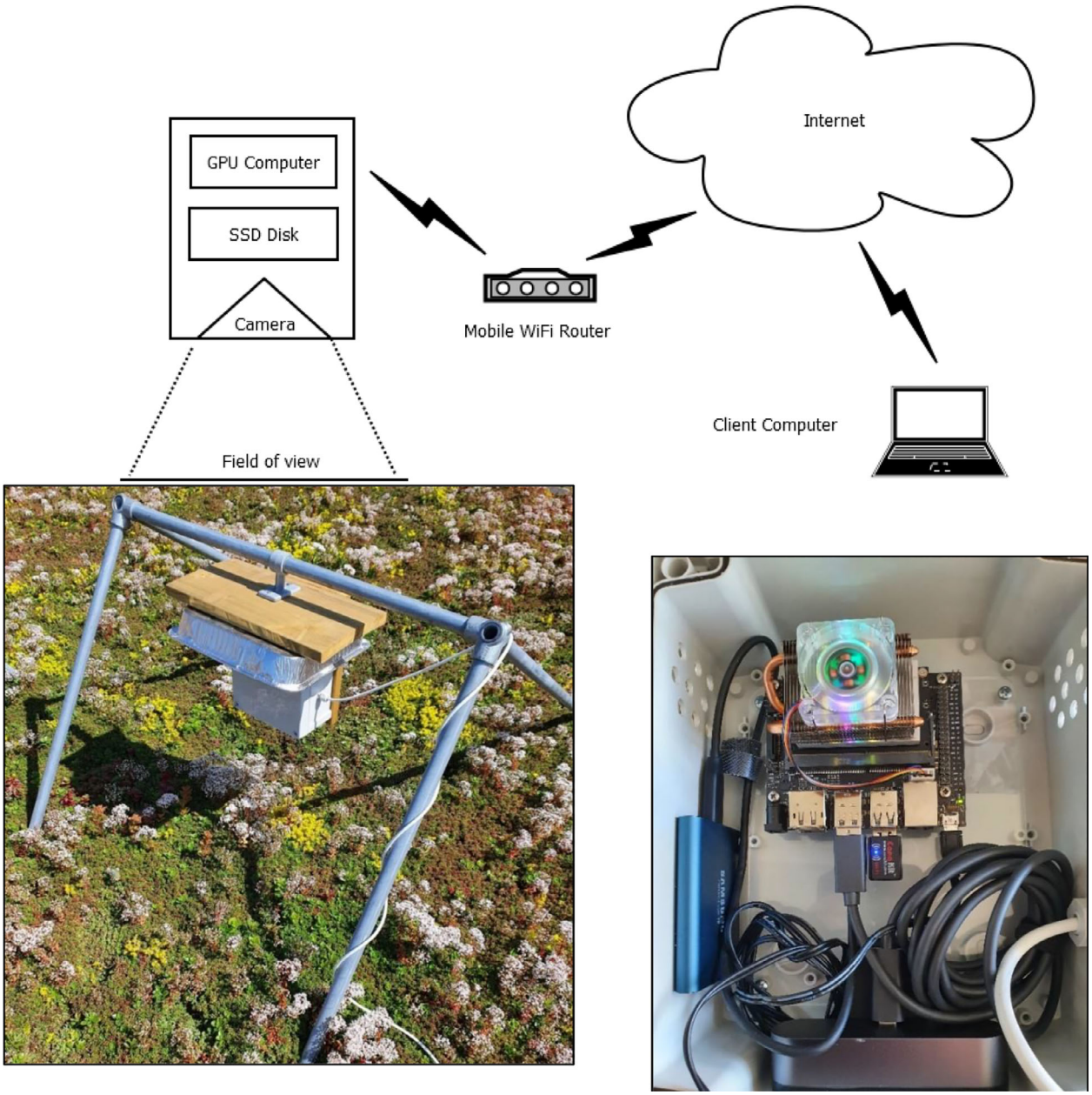


Figure 1. The portable computer vision system with internet access. The system consists of an intelligent camera and a mobile WiFi router connected to the internet. The intelligent camera with GPU computer (Jetson Nano), cooling, USB WiFi, Web camera, and SSD disk is built to detect and classify insects in real-time and only information about detections is transferred to the client computer via 4G internet. On daily basis, the detections are analysed to monitor tracks of the identified insect species on the client computer.

during full operation. In our study, we supplied power to the system through a mains line (230–240 V AC) connected to a 5 V/2.5 A transformer. Alternatively, a 12 V 55 Ah lead battery, DC-DC converter and a 60 Wp solar panel with charger should be sufficient for powering the system during the entire summer period (May–September in Denmark) for locations without power.

Software solution

The Jetson Nano computer executes two programs, which in parallel handle imaging, insect detection, and storing of the resulting data. The camera system continuously records images using a fixed sample interval with a speed of 0.33 fps. In parallel, the system runs a trained deep

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learning YOLOv3 (Redmon & Farhadi, 2018) model, which performs real-time detection and classification and records information on the position and species identity of detected insects in each image. The YOLOv3 framework used in this study was ported and rewritten to record information on object detections for each day as text files. YOLOv3 was able to process images with an inference time of approximately 0.5 fps running on the Jetson Nano computer. The inference should be faster than the frame rate of the camera to process data in real-time. Only the information on detections was sent via the internet while the images were stored on the SSD disk in the camera system to reduce the demand for data transfer. In addition, the tracking algorithm identified sequences of images of the same individual insect based on the detection and classification data from the image sequences. The insect tracks were computed from detection data accessed via the internet on a daily basis.

The motion program (Motion, 2021) installed on the Jetson Nano was used to continuously capture time-lapse images at a frame rate of 0.33 fps with a resolution of 1920 × 1080 pixels. This program could also have been configured to trigger whenever movement is detected in the camera field of view. With continuous time-lapse recording or constant movement, the camera would thus record 1200 images per hour, limited by the processing of each image. The YOLOv3 deep learning model was running in parallel with the motion program to perform inference on the captured recordings. To conserve memory, only images containing insect detections with a YOLOv3 confidence threshold above 10% were saved on the SSD drive. Alternatively, a sample or all images could be saved for later analysis.

We developed a computer vision algorithm referred to as *Insect Classification and Tracking* (ICT), which was able to track and count the number of insects and identify known species by deep learning. (Open source code: https://github.com/kimbjerge/insectTracking) The algorithm produced data of individual insect tracks and their species identity. For every individual insect track detected, the start time, duration, mean size of the bounding box defined by the insect, and species identity were recorded.

The ICT algorithm consists of five sequential steps (Fig. 2). The first two steps were performed on site using the embedded Jetson Nano computer, while the last three steps were done in the lab. The first part, performed on site (Capture images and Detect objects) involved capturing time-lapse images and detecting individual insect in the images using our trained CNN model. The location of each insect in the image was estimated in the second step (Detect objects). This first part is described in more detail in Section 2.2.1.

The second part was performed remotely. In this part, the first step (Filtering) filtered out objects where no motion was detected since many of them were not insects. The next step (Tracking) tracked moving insects in the image sequence. Tracking ensured that each individual insect was counted only once during each visit. Every image in each insect track was assigned to one of the known insect classes, and in case of disagreement across the images in a given track, the insect identity for the track was determined by majority vote. The final step (Statistics) stored data on timing, duration, and total length of each track for each individual insect detected and tracked by the algorithm along with the species identity in a text file for further study. Further details of the second part of the pipeline are given in Section 2.2.2.

Insect species classification (On site)

We used the CNN darknet53 (YOLOv3) with a total of 53 layers, which has a high mean Average Precision (mAP) on the COCO dataset of 57.9% (Lin et al., 2014). This accuracy is comparable to other state-of-the-art deep learning networks like RetinaNet (Liu et al., 2017), ResNet50 (Huang & Wang, 2017), and Faster R-CNN (Girshick, 2015), but with faster processing time. Darknet53 uses three residual skip connections that make detections at three different scales. The darknet53 is better at detecting small objects than the previous YOLOv2 (Ju et al., 2019). The predicted feature map of darknet53 contains bounding box coordinates and a confidence score for each detected object class. We chose an image resolution of 608 × 608 for inference with YOLOv3.

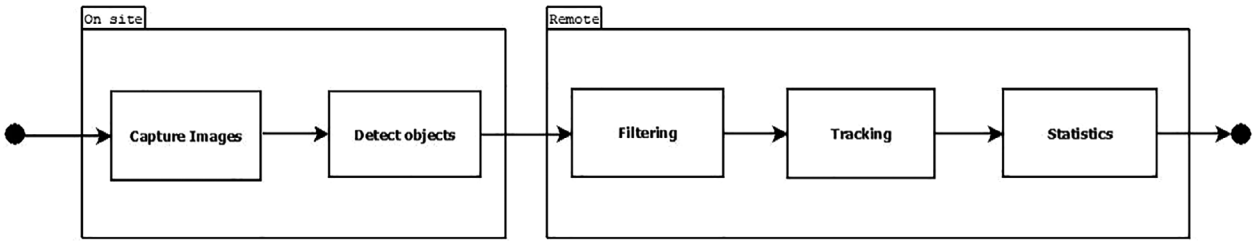


Figure 2. The processing pipeline of the on site and remote image processing algorithms to track and count the number of insect tracks. A trained convolutional neural network (CNN) was used for the insect species classification.

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We explored different combinations of image size, number of training iterations, and kernel filter size to find the optimal darknet53 network parameters to classify eight insect species. For the final model, we used the default settings of YOLOv3 but trained with an image resolution of 832 × 832 pixels and validated on images with 608 × 608 pixels. This approach gave better results than training on images with 608 × 608 pixels. The higher resolution of the training images also resulted in a higher accuracy than the recommended modification of darknet53 (YOLOv3) for small objects that specifies how to change the skip connection layers to detect small objects on different scales.

The training, validation, and test dataset for the deep learning model were based on images recorded during the summer 2019 on a mixed stand of plants in the genus *Sedum* (85%) (Høye et al., 2020) as well as a smaller set of images (15%) recorded on common heather *Calluna vulgaris* plants in Denmark. The complete training (including validation) and test datasets are shown in Table 1. The manually labelled images were selected to have high variation in both insects and plants from different camera locations and recorded at different times during the growing season with different species of blooming *Sedum* as well as common heather. All insects were annotated with bounding boxes. From the data, images of seven different frequently occurring species from different dates were selected, annotated, and identified to train the CNN model. In total, 2121 background images without insects and 5757 images with insects were used for training as seen in Table 1. Examples of individual insect species are shown in Figure 3. A separate class named 'Other arthropods' contains a mix of other insects like flies, ants, spiders, and insects with low-occurrence (247 flies, <60 of all other taxa) in the recordings mostly found on the location with heather plants. For the 'Other arthropods' and *Aglais urticae* class, we did not find sufficient images to create a test dataset. Table 1 shows an overview of the occurrence of all species in the annotated dataset for training and testing of the CNN algorithm.

Table 1. Number of insect species in the dataset used for training and test.

No.	Species	Train (Heather)	Train (*Sedum*)	Train total	Test (*Sedum*)
1	*Coccinellidae septempunctata*	22	374	396	97
2	*Apis mellifera*	365	1787	2152	266
3	*Bombus lapidarius*	0	305	305	50
4	*Bombus terrestris*	0	440	440	84
5	*Eupeodes corolla*	0	1147	1147	80
6	*Episyrphus balteatus*	68	705	773	169
7	*Aglais urticae*	13	123	136	0
8	Other arthropods	357	51	408	0
	Total	825	4932	5757	746
	Background	1223	898	2121	346

Most (85%) of the insects for training were annotated from the images with *Sedum* plants. The images of common heather (*Calluna vulgaris*) plants contributed with more data and improved the detection performance, particularly for backgrounds and the mixed insect class. The test dataset only contains images from the *Sedum* plants as used in our experiment.

Data augmentation was applied to all images using the default settings of YOLOv3. This operation provided more training data. Only the training dataset was used to train the YOLOv3 model and the number of training iterations was chosen based on when the average training loss was no longer decreasing. The selected YOLO3 model used 30 000 iterations, which gave a higher precision and recall on the test dataset than using less than 28 000 iterations. A higher number of training iterations would increase the risk of overfitting the model.

Table 2 shows an overview of precision and recall for the training and validation dataset using an Intersection over Union (IoU) overlap of 20%. Normally an IoU of 50% is used, but since the insects are very small in the images, the annotated and predicted bounding box may not overlap by 50%, even when the insect was detected correctly. To ensure that the predicted number of insects of each class approximated the actual number of insects (equal precision and recall) in each class in the images, we adjusted the confidence threshold (CT) of each class individually. In this way, there were just as many insects that the model did not find (false negatives) as there were predictions, which did not contain insects (false positives). However, another strategy could be to maximize recall by reducing the confidence threshold. The higher number of false positives found with such a strategy can be separated from true positives by manually sorting the predictions. The 'Other arthropods' class achieved the lowest precision and recall, probably because this class contains annotations with very different appearance. The precision, recall, and mean Average Precision (mAP) for validation uses the same CTs as for training. The validated YOLOv3 model had an average recall of 73%, indicating that 27% of the individuals in the validation dataset were missed. A precision of 72% was obtained indicating that 28% were wrongly classified including false detections in elements of the background. Especially the small hover flies *Episyrphus balteatus* and *Eupeodes corolla* were very difficult to separate. The low precision of 67% for *E. corolla* was due to 18 individuals wrongly classified by the algorithm, which should have been *E. balteatus*. A similar pattern was seen for *Bombus lapidarius* and *B. terrestris*. An average mAP of 87% was high compared to

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Figure 3. Examples of the insect species that were observed and labelled for training of the YOLOv3 model. The numbers refer to the species listed in Table 1.

Table 2. Precision, recall, and confidence threshold (CT) on the training and test dataset.

No.	Species	Precision (Training)	Recall (Training)	CT (%)	Precision (Test)	Recall (Test)	mAP (Test)
1	*Coccinellidae septempunctata*	0.97	0.94	30	0.72	0.93	0.84
2	*Apis mellifera*	0.95	0.95	30	0.67	0.93	0.78
3	*Bombus lapidarius*	0.98	0.95	30	0.87	0.78	0.97
4	*Bombus terrestris*	1.00	0.93	30	0.92	0.73	0.95
5	*Eupeodes corolla*	0.92	0.89	20	0.67	0.55	0.77
6	*Episyrphus balteatus*	0.93	0.91	15	0.87	0.36	0.91
7	*Aglais urticae*	0.93	0.95	35	n/a	n/a	n/a
8	Other arthropods	0.91	0.69	10	n/a	n/a	n/a
	Average	0.95	0.91	24	0.72	0.73	0.87

Mean Average Precision (mAP) is only measured for the test dataset.

58% for the mAP on the COCO dataset. However, the COCO dataset has many more classes (80) compared to the eight classes in our study and more conservative IoU thresholds.

Filtering and insects tracking (Remote)

In a previous study using time-lapse cameras, we observed that many false insect detections were made in specific structures of plants where no insects were found (Høye et al., 2020). With a higher imaging frequency, precision and recall could increase. With more images at a higher sampling rate, it can be assumed that more insects are detected, which would increase the recall. We introduced a filter to remove stationary false detections in the background image. Specifically, we filtered all detections for which the same position was detected within the past 5 min. This filter removed 87% of all detections assumed as false detections in the period of the experiment with a total 161 468 of detections. This high number of false detections is caused by the model erroneously detecting parts of plants as insects in the background image.

We used the multiple object tracking algorithm developed for moth tracking (Bjerge et al., 2021) to prevent multiple counting of the same insect in a sequence of images. The position and size of each insect were estimated for every single frame, and tracking was solved by finding the optimal assignment of individual insects in two consecutive images. We chose the Hungarian Algorithm (Munkres, 1957) for finding the optimal assignment for a given cost matrix. In this application, the cost matrix should represent how likely it was that an insect in the previous image had moved to a given position in the current image. The cost function was defined as a weighted cost of distance and area of matching bounding boxes in the previous and current image. For more details, see Appendix, Tracking Algorithm.

Testing the insect monitoring system

An experiment to test the functionality and stability of the system was conducted in the city of Aarhus, Denmark during the period 25 June to 29 September 2020, where one system was active daily from 6:30 am to 9:00 pm. The automated system was located on top of a green roof comprised of 13 species of *Sedum* plants (Høye et al., 2020). The camera was installed 30 cm above the plant and flowers with a field of view of 46 × 27 cm, which is a sufficiently large area while maintaining image quality for identifying insect species. The camera used automatic exposure to handle variation in light conditions related to direct sun, clouds, and shadows. Focus was manually adjusted for optimal image sharpness. The captured images were stored in JPG format with a resolution of 1920 × 1080 pixels. Based on the software described

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above, a summary of the detected insect tracks was generated. Each track was recorded with date and time for start and end of the visit. The predicted species of the insect with a confidence score and the number of images through which the insect had been tracked were also recorded. The confidence score was calculated as the ratio between the most frequent class and the total number of classifications in a track. Finally, for each track, the average bounding box size of the detections and the total Euclidean distance in pixels that the centre point of the bounding box had moved throughout a track was recorded. Several parameters (weighted cost for distance and area (W_{dist}, W_{area}), minimum cost to add detection to track, number of detections to define a valid track and time to terminated track) were defined and adjusted to filter noisy tracks such as insects only observed in one image. An insect should be detected in at least two images in a sequence for a track to be valid. More details on the tracking algorithm are described in Bjerge et al. (2021).

We calculated the number of tracks for each species in each day and compared this to the number of detections of each species in each day to quantify the prevalence of multiple counting of each individual. To illustrate how the track structure and duration varied depending on the type and abundance of flowers in the field of view of the camera, we further plotted the coordinates of each detection in each track for each of the insect species during each month of the season along with example images for the period. To examine taxa-specific variation in the duration of tracks (in seconds), we also computed density distributions of track durations for each taxa individually. Finally, we computed diurnal activity patterns for the most abundance taxa across the season.

To test the final insect monitoring system, a classification and tracking test dataset was produced based on ten movies from different days over the period of experiment in 2020 with variation in species and *Sedum* flowers. These movies were manually inspected to estimate the precision of the ICT algorithm. In total, 568 insect tracks were manually validated either as true, a false background or wrong species. None of the images from the experiment were used to train the YOLOv3 model. In this way, the ICT algorithm was tested on image data it had never seen before.

Results

During the experiment conducted in the period 25 June to 29 September 2020, the system only missed recordings from 4 days due to technical failure. The *Sedum* plant species flower at different times during the growing season, so there was a continuous availability of flowers from late May to September. Figure 4 left shows an example of an image taken by the camera with blooming *Sedum* plants and one honeybee. Figure 4 right shows a plot of all detected and classified insects after filtering from all images collected on the same day (15 August 2020). In the following sections, the results from the experiment in 2020 and the last stages of the algorithm concerning insect filtering and tracking (as described in Fig. 2) are presented.

Precision for insect tracking

A movie[1] was created for each day of observation marked with coloured insect tracks as shown in Figure 5 of two honeybees and of a bumblebee. For an insect track to be valid, at least two observations must be detected meaning that the insect was visible for the camera for at least three seconds. This filtering did remove many false positive background detections, which happened mostly in June, and start of July. In this period, white *Sedum* flowers were often detected as hoverflies *Eupeodes corolla* or *Episyrphus balteatus*.

Table 3 The number of images and detected tracks for each species and the number of True Positive (TP) tracks and estimated precision.

Temporal variation in insect abundance and duration of visits

The trained YOLOv3 model detected insects in 20 863 images after the stationary filtering (described in Section 2.2.2.) over the 98 days of recording, and the ICT algorithm summarized these into 2994 insect tracks. Some tracks were much longer than others leading to a divergence in the time series of individual detections and tracks (Fig. 6). In the bee and butterfly species, it was particular clear that the seasonal pattern of abundance was different when based on counts of individual detections and tracks. The largest number of insect tracks recorded in one day was 332 on 30 June 2020.

Figure 7 shows scatter plots with tracks of all insect detections grouped for bumblebees, honeybees, hoverflies and butterflies for each month, separately. *Bombus lapidarius* and *B. terrestris* are combined into one group for bumblebees. *Eupeodes corolla* and *Episyrphus balteatus* are plotted together for hoverflies. In the period 11 to 17 August, most insect tracks were detected with an average of 174 honeybee tracks per day ranging from 101 to 332 tracks.

[1]Example of insect tracking on YouTube: https://www.youtube.com/watch?v=tW-eEh0ASA4&t=157s.

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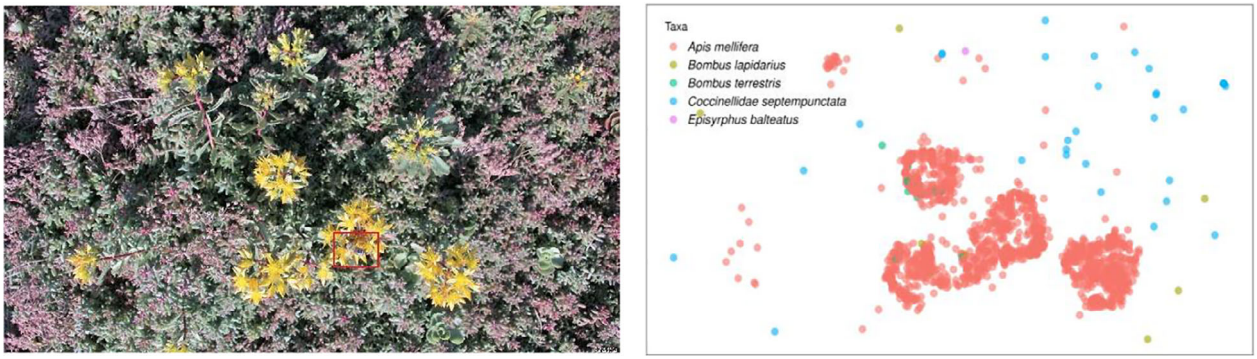


Figure 4. Image of *Sedum* plants with one honeybee (left) and detected insects in all images (right) from same day (15 August 2020).



Figure 5. Tracks of two honeybees (left) and track of one bumblebee (right).

Table 3. Number of images and tracks for each species in the test dataset detected by the algorithm.

Species	Images	Tracks	TP	Precision
Coccinellidae septempunctata	93	39	29	0.74
Apis mellifera	1700	348	342	0.98
Bombus lapidarius	446	62	56	0.90
Bombus terrestris	80	22	19	0.86
Eupeodes corolla	96	38	15	0.39
Episyrphus balteatus	64	34	18	0.53
Aglais urticae	145	22	22	1.00
Other arthropods	10	3	3	1.00
Total	2634	568	504	0.89

A True Positive (TP) is defined as an insect that was correctly classified in the entire track. Precision is the number of TP relative to the total number of tracks detected.

Figure 8 shows the distribution for the duration of honeybee residence times in the period with low and high abundance. The insect species all have the same exponential decay for the duration of visits with a residence time typically less than 1 min. The period 11–17 and 22 August, with high abundance of honeybees, more than 100 tracks per day was observed. The average duration of a visit was 33 s. and in the period with low abundance, a honeybee visit took on average 27 s.

Figure 9 shows that the diurnal activity patterns among insect taxa with the biggest difference between honeybees and hoverflies.

Discussion

In this study, we have presented an automated system for real-time insect monitoring. The system is widely applicable as a means to generate standardized monitoring data. We have shown that the system is capable of generating novel insights into the seasonal and diurnal dynamics for different flower visiting insect species. Remarkably, our fully automated insect monitoring camera system is able to perform tracking of individual insects in real time (Gilpin et al., 2017; Høye et al., 2021; Lortie et al., 2011; Pegoraro et al., 2020).

Motion detection in close range time-lapse photography is challenging as both the object of interest and

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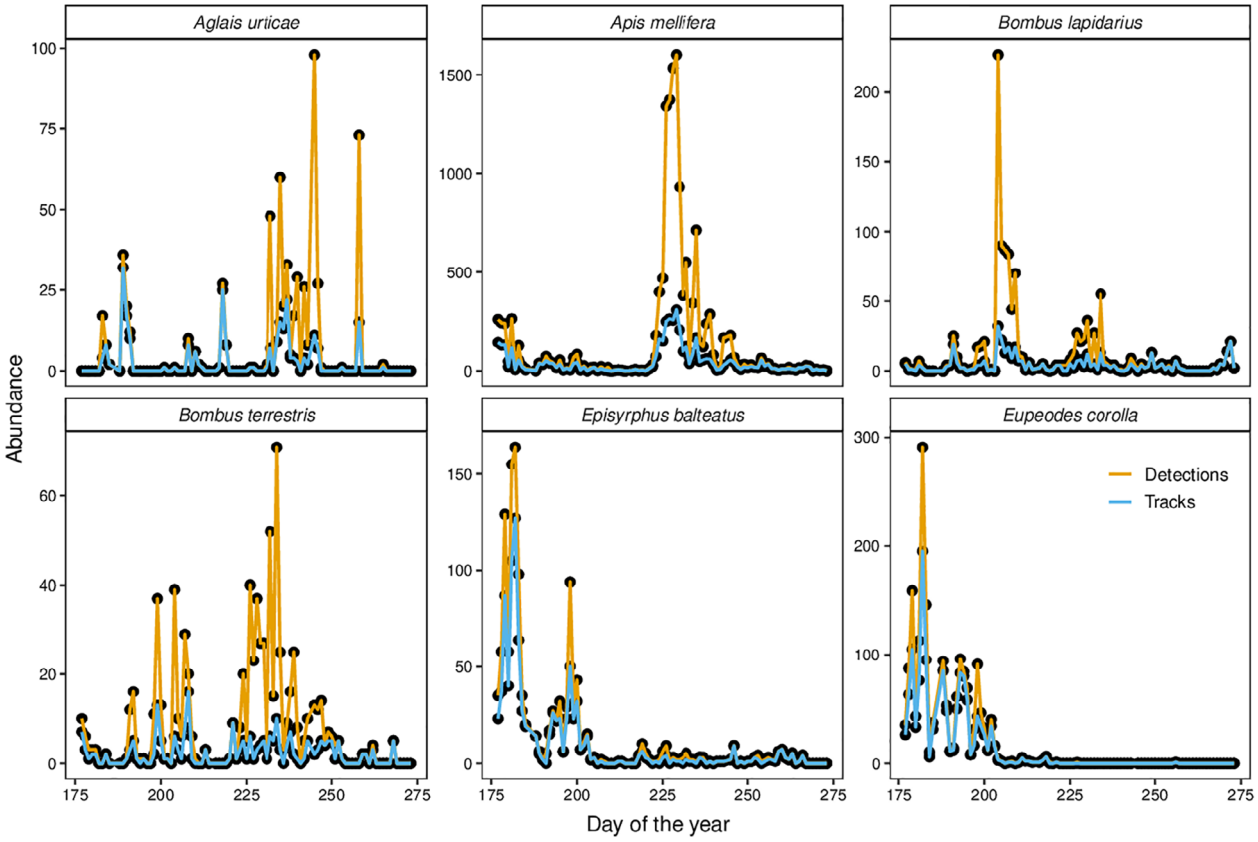


Figure 6. The seasonal dynamics of the number of individual detections and the number of tracks for each of the individual species detected.

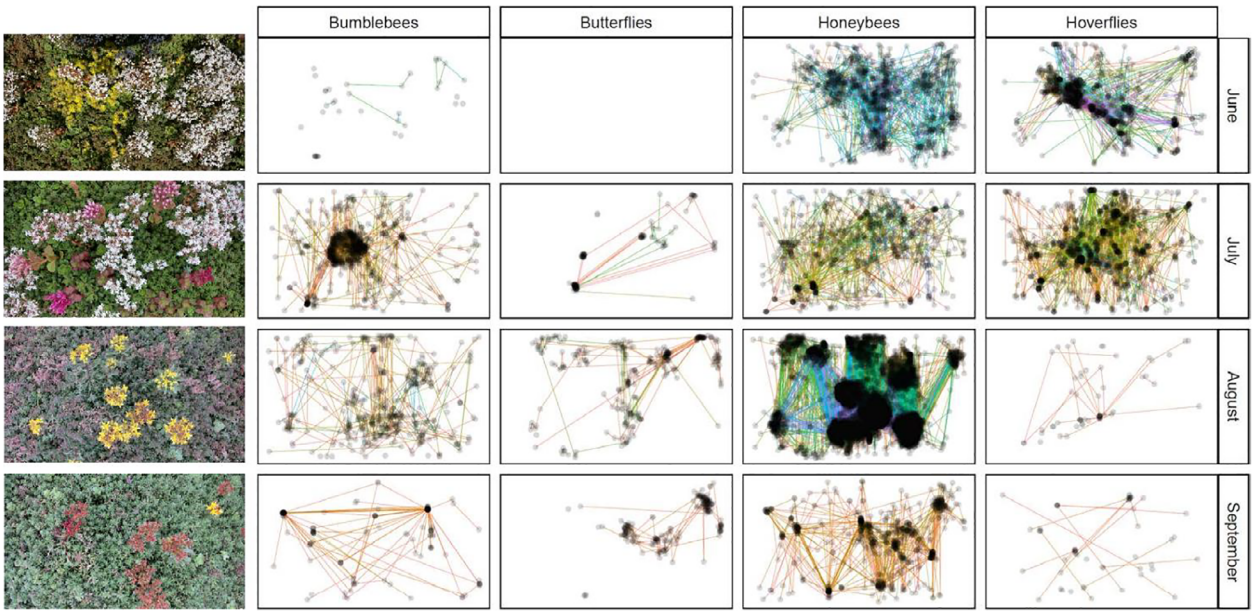


Figure 7. Detection coordinates grouped in four taxa for each month, separately. The individual detections are connected into tracks by line segments. Line segments of the same colour are from the same track and are plotted to illustrate how the detailed movement of individual flower visitors varies through the season and differ among insect taxa. Note that the camera system was moved occasionally during the season to maximize the recording of flower visiting insects.

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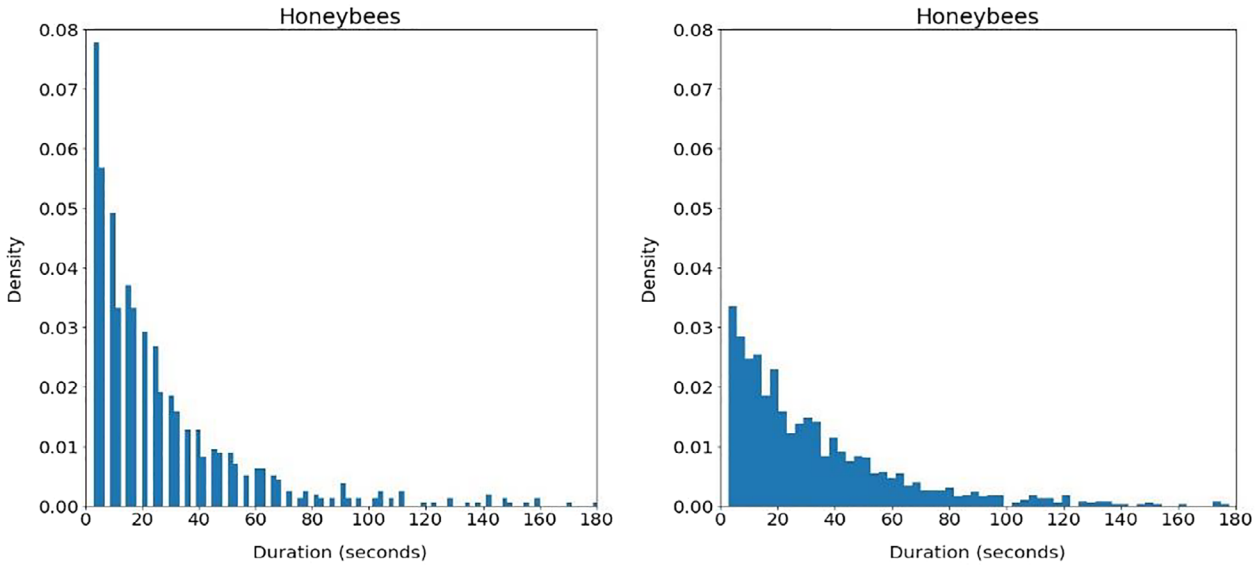


Figure 8. Visits in period with low abundance (left) and visits in period with high abundance (right).

elements of the background can move (Steen, 2017; Weinstein, 2018). Our system solves this important challenge for insect monitoring by applying CNN-enabled object detection during the recording process at an impressive frame rate. Our results demonstrate that this technical achievement has strong implications for the quality of the ecological data collected. We found a remarkable discrepancy between the abundance of detections and the abundance of tracks for some insect species. Since tracks are made by individuals, the number of tracks represents a more precise estimate of insect abundance than the number of individual detections, which could include multiple detections of the same individual. Furthermore, the collection of multiple images of each same specimen ensures a higher accuracy in the species identification process.

When object detection and classification are incorporated into the recording process, a high performance of the CNN is important. With our experiment, the system detected in total 2994 insect tracks over 98 days where the ICT algorithm measured an average precision of 89% in classifying the species correctly. The classification precision was poorest for hoverflies due to many false positives in which elements of the background were detected as insects. The most common species in the images was the honeybee *Apis mellifera*, which had a very high precision of 98%. The precision for the bumblebee species was intermediate and most errors were due to a mixing of the two most abundant species of bumblebees *B. terrestris* and *B. lapidarius*. Although the number of classes in which the system is able to discriminate among is relatively low, the high performance of the CNN supports our novel approach of incorporating computer vision and deep learning into the image recording process for insect monitoring.

The solution operated without interruptions over extended periods, even during rain, strong winds, and at high temperatures during the summer period. During our experiment, the second part of the algorithm was run after image recording was done, but could easily be moved to on site processing. By running the entire algorithm on the camera system, it would also be possible to present summary data in real time to the end user of a specific unit or to combine summary data from multiple units at regional or even global scale to a wider community. Once the system is launched, it requires no maintenance, and the communication via the internet provides an opportunity to monitor status of each system. To reduce costs, the camera can be replaced by a cheaper C920 model (Logitech, 2020), but careful consideration should be taken if this is done at the expense of image quality to ensure sufficient image quality for specific applications. Further, the SDD hard drive could be replaced by a SD card to reduce the overall cost of the hardware from approximately $600 to $425 with grid power. A Solar panel, a battery and a charger add an additional cost of $425 for locations without power. As such, we argue that the solution is tested, stable, and reasonably priced and thus ready for large-scale deployment for insect monitoring across larger regions.

The number of insect visits we recorded varied among species and over the study period. Most remarkably, we did not find clear evidence of a confined seasonal peak in activity for bumblebees, hoverflies and butterflies, while

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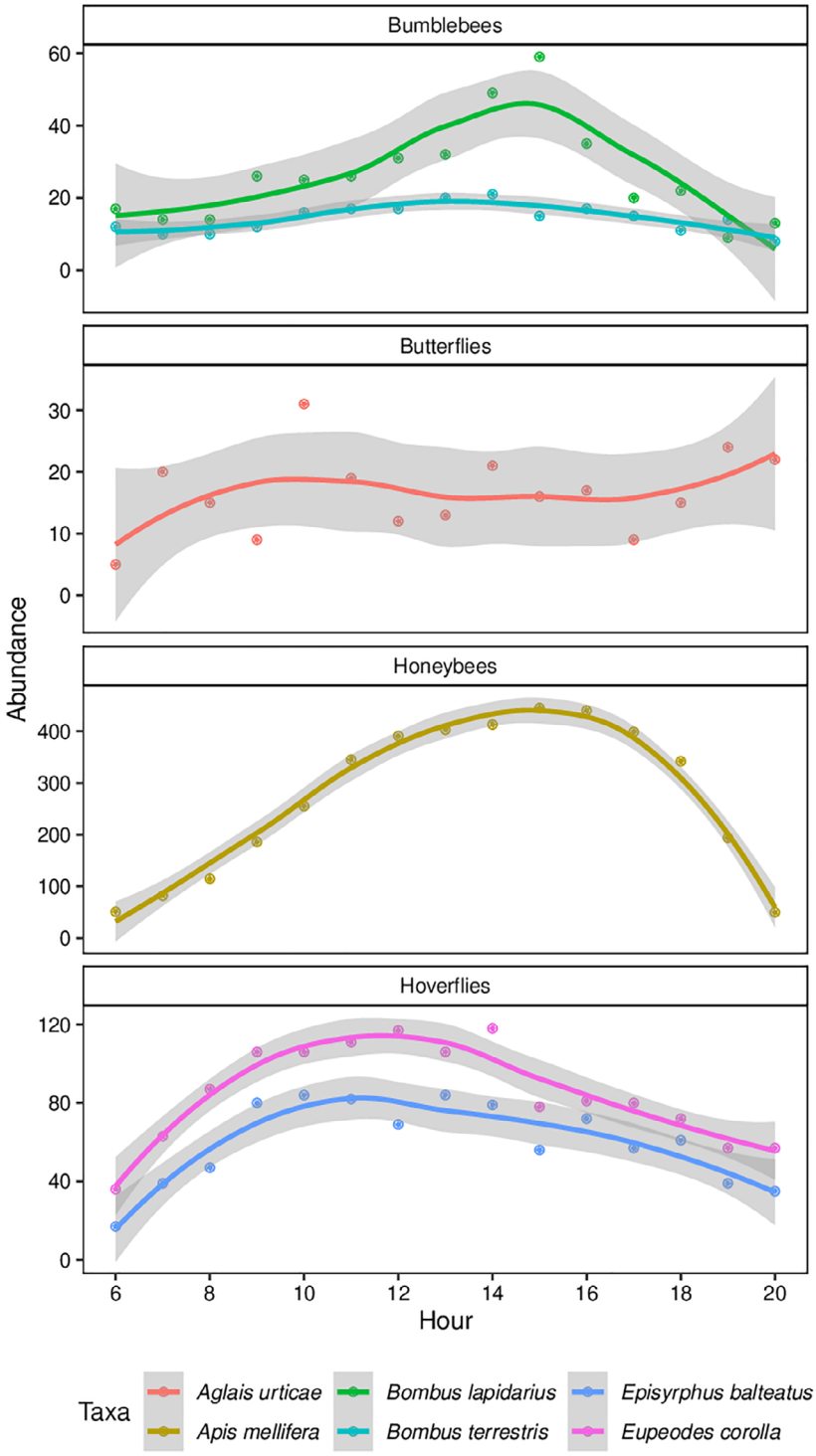


Figure 9. Diurnal activity patterns summed across 98 days of observation varied substantially among insect taxa with the biggest difference between honeybees and hoverflies.

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honeybees were most abundant in mid-August. Nevertheless, our data do map clear differences in the diel activity of different insect taxa. Specifically, the hoverflies seem to be active earlier during the day than the bees and the butterflies. Such information have implications for understanding the risk of competition among cultivated and wild pollinators (Wignall et al., 2020; Wojcik et al., 2018). Due to the high temporal resolution of the data recorded by our system, there is also great potential for using the system to provide a better understanding of the weather-related drivers of insect visitation rates (Zoller et al., 2020). Finally, we also observed that the duration of honeybee visits is longer in periods of high honeybee abundance. A honeybee used on average 22% more time during a visit with a 28% longer track length in the period with high honeybee abundance. While this could be evidence of floral resource depletion, as each bee may spend more time foraging during periods of high bee abundance, this could also be related to the amount of nectar and pollen produced by each species of flowering plants present at different parts of the season. It will be important to match insect detections with the amount and composition of floral resources in the scene of the camera during future large-scale comparative studies. For a wider application of the system, it will be important to test the system at different locations with a wider range of insect species. In this context, the CNN classification algorithm can be improved by adding more annotations to the training data set for the same species, but also for additional insect species. In any case, the proposed computer vision system and ICT algorithm are promising as an automated system for insect observations and complement existing systems for monitoring nocturnal insects (Bjerge et al., 2021).

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix S1. Tracking algorithm.

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